

E-Commerce Recommendation Platform

Using Apache Spark, GraphFrames, VADER Sentiment Analysis, and Ray
Serve

CSCI-GA.2437 Big Data Application Development - Spring 2026
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April 2026

Abstract

We present an end-to-end e-commerce recommendation platform built on the Amazon 2023 Appliances dataset comprising over 2.1 million product reviews. The system processes raw data through a distributed Apache Spark pipeline covering profiling, cleaning, joining, VADER sentiment analysis, and Spark SQL analytics. A collaborative filtering model using Alternating Least Squares (ALS) is trained following systematic hyperparameter optimization across 27 parameter combinations, yielding a final model with Test RMSE of 1.8832, a 16% improvement over the baseline. Product recommendations are further enriched using graph-based PageRank scores computed with GraphFrames on a co-purchase graph of 77,633 product vertices and 391,037 weighted edges. The entire inference pipeline is served in real-time via Ray Serve, achieving approximately 45ms latency per user and supporting batch simulation of 10,000 users simultaneously. A web-based frontend provides an Amazon-style interface for personalized recommendations and large-scale user behavior simulation.

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1 Introduction

Product recommendation is central to the modern e-commerce experience. As product catalogs grow to millions of items and user bases reach hundreds of millions, the fundamental challenge shifts from finding relevant products to efficiently computing relevance at scale. A single user’s preferences must be matched against the entire product catalog in milliseconds, while the model itself must be trained on billions of user-product interactions.

This project addresses that challenge by building a full-stack recommendation system that demonstrates every stage of a real-world Big Data pipeline. Starting from raw compressed JSON reviews sourced from the Amazon 2023 corpus, we implement distributed data profiling and cleaning, sentiment-enriched analytics, collaborative filtering with systematic hyperparameter optimization, graph-based reranking via PageRank, and finally real-time model serving using Ray Serve.

The Amazon Appliances subset was selected as it provides a meaningful scale that genuinely requires distributed processing: over 2.1 million reviews, 94,000 products, and 1.8 million unique users. The pipeline is designed to scale horizontally to the full multi-category Amazon dataset as future work.

1.1 Why Big Data?

The scale of this problem cannot be addressed on a single machine:

- The raw reviews data is 257.7 MB compressed, expanding to several GB uncompressed, too large for in-memory processing on a laptop
- ALS matrix factorization on a $1.6\text{M} \times 77\text{K}$ interaction matrix requires distributed linear algebra
- VADER sentiment analysis across 1.87 million text reviews using single-threaded Python would take hours; distributed with Spark `mapPartitions` it completes in minutes
- Generating recommendations for all 1.3 million users simultaneously requires distributed matrix-vector multiplication
- GraphFrames PageRank on 391,037 edges cannot run efficiently on a single node

1.2 Contributions

- Complete end-to-end distributed pipeline on the Amazon 2023 dataset with the updated schema
- Systematic HPO over 27 ALS parameter combinations using distributed cross-validation
- Graph-based reranking combining ALS collaborative filtering with GraphFrames PageRank
- Real-time Ray Serve deployment with three chained deployments and approximately 45ms per-user latency
- Web application with personalized recommendations and 10,000-user simulation

2 Dataset Description

2.1 Source

The dataset is sourced from the **UCSD Amazon 2023 Product Reviews** corpus [1], available at <https://amazon-reviews-2023.github.io/>. This is a substantially revised version of the widely used McAuley Lab Amazon datasets, incorporating reviews through September 2023. We use the **Appliances** category subset, which covers home appliances, replacement parts, water filters, ice makers, and related accessories.

2.2 Dataset Statistics

Table 1: Raw Dataset Overview

Dataset	HDFS Path	Size / Rows
Reviews	.../raw/appliances_reviews.json.gz	257.7 MB / 2,128,605 rows
Metadata	.../raw/appliances_metadata.json.gz	63.3 MB / 94,327 rows

2.3 Reviews Schema

The Amazon 2023 schema differs significantly from the 2018 version. Several critical changes affected our processing pipeline:

Table 2: Amazon 2023 Reviews Schema

Field	Type	Notes
user_id	String	Unique reviewer ID (renamed from <code>reviewerID</code>)
parent_asin	String	Parent product ASIN, canonical identifier
asin	String	Specific variant ASIN
rating	Float	Star rating 1.0–5.0 (renamed from <code>overall</code>)
text	String	Review body (renamed from <code>reviewText</code>)
helpful_vote	Long	Pre-aggregated helpful votes
verified_purchase	Boolean	Verified purchase flag
timestamp	Long	Milliseconds since epoch (changed from seconds in 2018)
title	String	Review title
images	Array	Review image URLs

2.4 Metadata Schema

Table 3: Amazon 2023 Metadata Schema

Field	Type	Notes
parent_asin	String	Canonical product identifier
title	String	Product title
store	String	Brand/seller name
categories	Array	Nested category hierarchy
price	Float	Listed price (50.46% null)
average_rating	Float	Aggregate rating
rating_number	Long	Total rating count
images	Struct	Nested image URLs (hi_res, large, thumb)
bought_together	Array	Co-purchase ASINs (100% null in Appliances)
features	Array	Bullet-point product features

3 System Architecture

The platform is built as a sequential pipeline of eleven stages, each implemented as a separate PySpark script and running on the NYU Google Cloud Dataproc cluster (Spark 3.5.3, Hadoop 3.3.6).

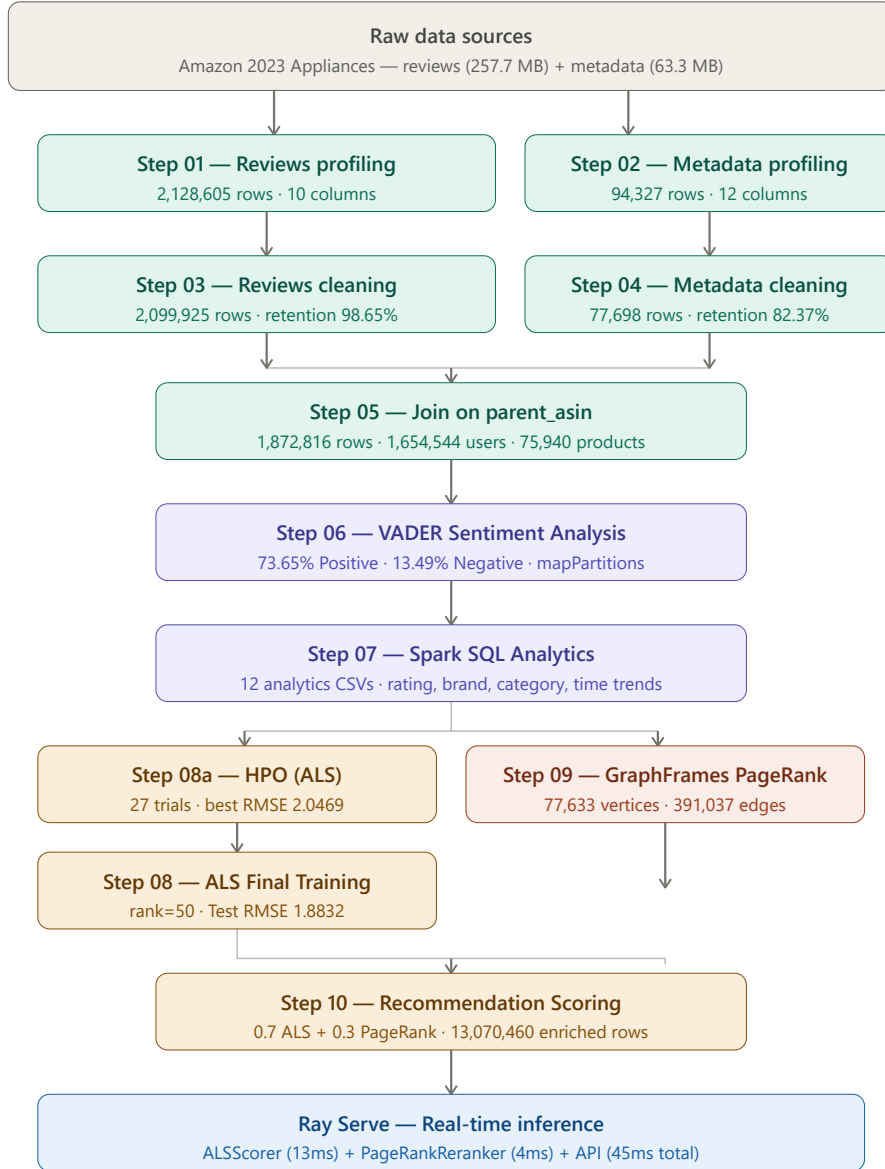


Figure 1: End-to-end pipeline architecture. Raw data flows through profiling, cleaning, joining, sentiment analysis, analytics, HPO, ALS training, GraphFrames PageRank, recommendation scoring, and Ray Serve real-time inference.

Table 4: Pipeline Stages and Output Scale

Step	Script	Description	Output Rows
01	01_profiling_reviews.py	Reviews profiling	statistics only
02	02_profiling_metadata.py	Metadata profiling	statistics only
03	03_cleaning_reviews.py	Reviews cleaning	2,099,925
04	04_cleaning_metadata.py	Metadata cleaning	77,698
05	05_join.py	Join reviews + metadata	1,872,816
06	06_sentiment_analysis.py	VADER sentiment	1,872,816
07	07_spark_sql_analytics.py	Spark SQL analytics	12 CSVs
08a	08a_hpo.py	ALS hyperparameter optimization	27 trial JSONs
08	08_als_training.py	Final ALS model	model artifacts
09	09_graphframes_pagerank.py	GraphFrames PageRank	77,633 scores
10	10_recommendation_scoring.py	Final recommendations	13,070,460

3.1 Technology Stack

Table 5: Technology Stack

Component	Technology
Cluster	NYU Google Cloud Dataproc (3 nodes)
Distributed Computing	Apache Spark 3.5.3 (DataFrame API + Spark SQL)
Graph Processing	GraphFrames 0.8.3
Sentiment Analysis	VADER via <code>vaderSentiment</code>
Recommendation Model	ALS (MLlib)
Model Serving	Ray Serve 2.x + Starlette
Storage	HDFS (Hadoop 3.3.6) + Parquet
Frontend	HTML5 + JavaScript + Chart.js
Tunnel	ngrok (free tier)
Language	Python 3.11 (PySpark)

4 Data Profiling

4.1 Reviews Profiling (Step 01)

The reviews profiling script computed null counts, rating distributions, temporal coverage, duplicate detection, and text quality metrics across all 2,128,605 raw rows.

Table 6: Key Reviews Profiling Findings

Metric	Value
Total rows	2,128,605
Unique users	1,808,565
Unique parent ASINs	93,671
Average rating	4.22
5-star reviews	69.82%
1-star reviews	11.77%
Verified purchases	95.84%
Reviews with no alphabetic text	4,141
Reviews containing HTML tags	159,509
Exact duplicate rows	22,657
<code>asin</code> $\neq$ <code>parent_asin</code>	34.69% of reviews
Timestamp format	Milliseconds (confirmed)
Oldest review	2000-10-23
Newest review	2023-09-12

The rating distribution shows a strong positive skew: 69.82% of reviews are 5-star, reflecting the well-known positivity bias in voluntary review systems. The 34.69% discrepancy between `asin` and `parent_asin` is a key 2023-specific finding. Using raw ASINs for the recommendation model would fragment the same product across multiple identifiers, degrading model quality. We use `parent_asin` throughout as the canonical product identifier.

4.2 Metadata Profiling (Step 02)

Table 7: Key Metadata Profiling Findings

Metric	Value
Total rows	94,327
Unique parent ASINs	93,671
Unique brands (stores)	12,568
Null price	50.46%
Avg price (where available)	\$86.48
Max price	\$21,095.62
Products with hi-res image	82.38%
Products with features	82.46%
<code>bought_together</code> null rate	100%
Avg category depth	3.32 levels
Duplicate titles	3,578

The most notable finding is the 100% null rate on the `bought_together` field. In earlier Amazon dataset versions this field provided co-purchase edges directly; its complete absence in the Appliances 2023 data required a workaround: we construct the co-purchase

graph from review co-occurrence instead. The 50.46% null price rate also informed our downstream UI design – products without prices are displayed as “Price N/A” rather than imputed.

5 Data Cleaning

5.1 Reviews Cleaning (Step 03)

Table 8: Reviews Cleaning Pipeline

Stage	Rows Removed	Rows Remaining	Notes
Raw dataset	0	2,128,605	starting point
Null <code>user_id</code> / <code>asin</code>	0	2,128,605	Data already clean
Invalid ratings (out of 1–5)	0	2,128,605	All ratings valid
No alphabetic text	4,141	2,124,464	Symbols/whitespace only
HTML tag stripping	0 removed	2,124,464	159,833 rows modified
Deduplication (<code>user</code> , <code>parent_asin</code>)	24,539	2,099,925	Keep most recent
Final	28,680 (1.35%)	2,099,925	Retention: 98.65%

Three operations deserve detail. First, HTML stripping was applied selectively – only reviews matching a regex pattern for HTML tags or entities were processed through a Python UDF, avoiding unnecessary overhead on clean text. Second, deduplication uses `parent_asin` so that a user who reviewed two variants of the same product is treated as a single review; the most recent review by timestamp is kept. Third, the timestamp normalization divides by 1000 to convert from milliseconds to seconds, a change required by the 2023 schema.

```

1 df = df.withColumn("timestamp",
2   (F.col("timestamp") / 1000).cast(LongType()))
3 ).withColumn("review_date",
4   F.from_unixtime(F.col("timestamp")).cast(DateType()))
5 )

```

Listing 1: Timestamp normalization – 2023 schema requires divide by 1000

5.2 Metadata Cleaning (Step 04)

Table 9: Metadata Cleaning Pipeline

Stage	Rows Removed	Rows Remaining	Notes
Raw dataset	0	94,327	starting point
Null <code>title</code>	9	94,318	
Missing hi-res image	16,620	77,698	Products without images excluded
Duplicate <code>parent_asin</code>	0	77,698	No duplicates found
Final	16,629 (17.63%)	77,698	Retention: 82.37%

The metadata cleaning script also performs: brand normalization via `F.initcap()` to resolve case variants (e.g., “FRIGIDAIRE” becomes “Frigidaire”); category extraction from the nested `categories` array-of-arrays into flat `main_category` and `subcategory` fields; and image URL extraction preferring `hi_res` over `large` over `thumb`. A notable schema workaround was also required – the `details` field contained duplicate keys in some records, causing Spark’s default JSON inference to fail. An explicit `StructType` schema definition was required to load the file correctly.

6 Data Integration

Reviews and metadata are joined on `parent_asin` using an inner join strategy. This ensures only reviews with valid product metadata enter the model training pipeline.

Table 10: Join Results

Metric	Value
Reviews input	2,099,925
Metadata products	77,698
Reviews with no metadata match	227,109 (10.82%)
Joined rows	1,872,816
Retention rate	89.18%
Unique users in joined data	1,654,544
Unique products in joined data	75,940

The 10.82% of reviews that did not match metadata correspond to products present in reviews but absent from the metadata dump – this is typical of Amazon datasets where product listings are occasionally removed. The joined dataset covers the top three main categories by review volume: Tools & Home Improvement (36.41%), Appliances (28.15%), and Amazon Home (27.27%). The top subcategories by volume are Water Filters (308K), Replacement Parts (243K), and Reusable Filters (176K).

7 Sentiment Analysis

7.1 Approach

Sentiment analysis is applied to all 1,872,816 review texts using the VADER (Valence Aware Dictionary and sEntiment Reasoner) lexicon [3]. VADER is a rule-based model specifically designed for social media and review text, making it well suited to Amazon reviews without requiring model training.

Each review’s compound score is classified as: Positive ($\geq +0.05$), Negative (≤ -0.05), or Neutral (between the two thresholds).

A key engineering challenge was distribution: YARN worker nodes did not have `vaderSentiment` installed. The package was zipped and shipped to all workers via Spark’s `--archives` mechanism. Additionally, to avoid the overhead of creating a new `SentimentIntensityAnalyzer` for every row, the `mapPartitions` pattern is used – one analyzer instance is created per partition.

```
1 def analyze_partition(rows):
2     from vaderSentiment.vaderSentiment import
3         SentimentIntensityAnalyzer
4     analyzer = SentimentIntensityAnalyzer() # once per partition
5     for row in rows:
6         text = row["review_text"]
7         if text is None or str(text).strip() == "":
8             score, label = 0.0, "Neutral"
9         else:
10            score = float(
11                analyzer.polarity_scores(str(text))["compound"]
12            )
13            if score >= 0.05: label = "Positive"
14            elif score <= -0.05: label = "Negative"
15            else: label = "Neutral"
16        yield (*row, score, label)
```

Listing 2: Efficient VADER scoring via mapPartitions

7.2 Results

Table 11: Sentiment Distribution (1,872,816 Reviews)

Label	Count	Percentage
Positive	1,379,369	73.65%
Neutral	240,784	12.86%
Negative	252,663	13.49%
Total	1,872,816	100%

The sentiment results show strong alignment with star ratings, as illustrated in Figure 2. One-star reviews average a compound score of -0.131 , while five-star reviews average

+0.551 – a monotonic progression that validates VADER’s suitability for this domain.

A disagreement analysis reveals that only 9.17% of reviews show mismatch between rating and sentiment: 4.23% have high ratings but negative sentiment, and 4.94% have low ratings but positive sentiment (Figure 3).

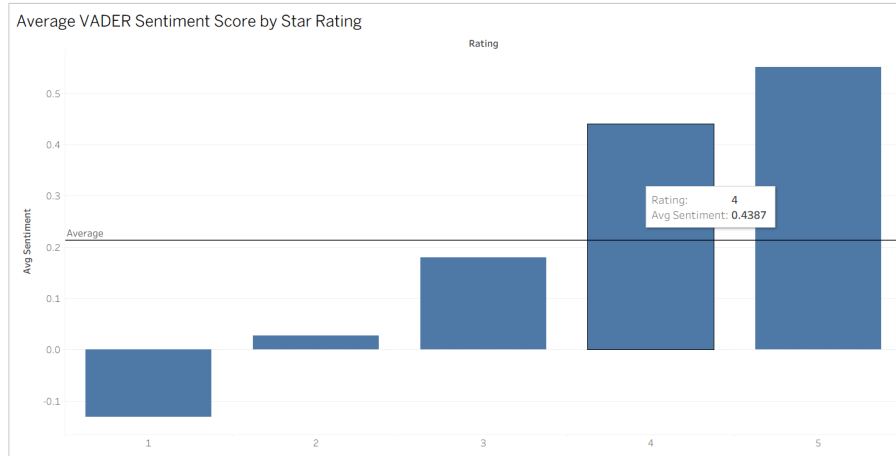


Figure 2: Average VADER Sentiment Score by Star Rating. The horizontal reference line marks the overall average. A perfect monotonic increase validates sentiment-rating alignment.

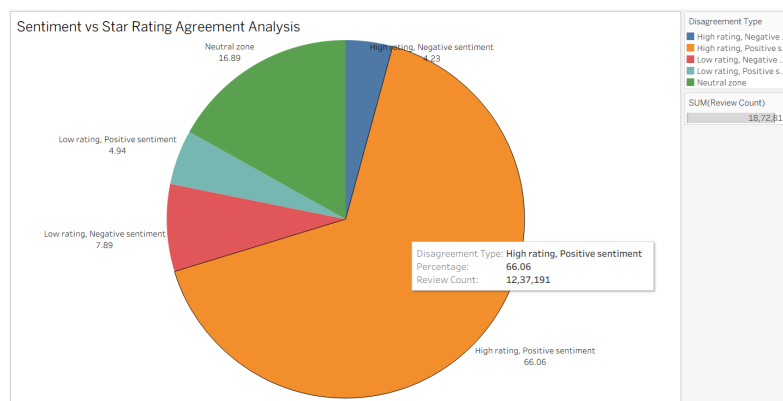


Figure 3: Sentiment vs Star Rating Agreement. 66.06% of reviews show consistent high rating and positive sentiment. Only 9.17% show disagreement between rating and sentiment.

8 Spark SQL Analytics

The sentiment-enriched joined dataset is registered as a Spark SQL temporary view and queried across twelve analytical dimensions. This section presents each analysis with quantitative findings and interpretation of the business insights they surface.

8.1 Rating Distribution

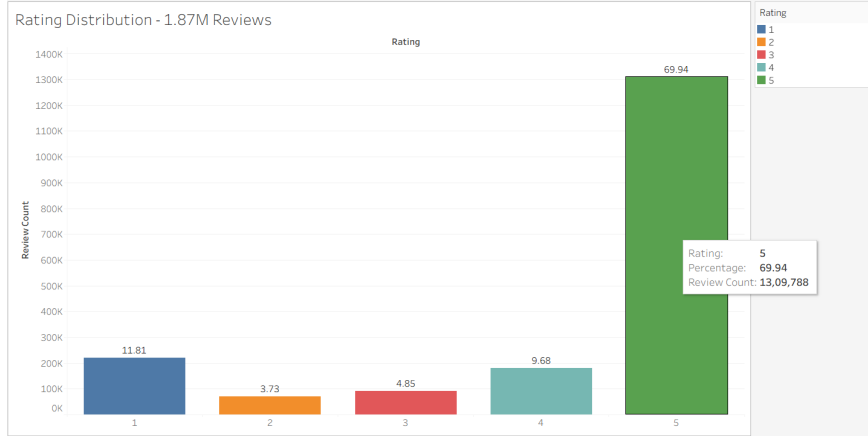


Figure 4: Review Rating Distribution across 1.87M reviews. The distribution shows extreme positive skew with 69.94% five-star reviews.

Table 12: Rating Distribution with Sentiment Alignment

Stars	Count	Percentage	Avg Sentiment	Avg Helpful Votes
1	221,095	11.81%	−0.131	1.80
2	69,829	3.73%	+0.027	1.15
3	90,905	4.85%	+0.181	1.15
4	181,199	9.68%	+0.439	1.15
5	1,309,788	69.94%	+0.551	0.72

The distribution reveals two important insights. First, the J-shaped curve (most reviews either 1-star or 5-star) is typical of Amazon reviews and reflects a self-selection bias: customers are most motivated to review when they are either very satisfied or very dissatisfied. Second, 1-star reviews receive on average 1.80 helpful votes compared to 0.72 for 5-star reviews, suggesting that critical reviews are considered more informative by other buyers – they help people avoid a poor purchase. The near-perfect monotonic alignment between star rating and VADER compound score (from -0.131 at 1-star to $+0.551$ at 5-star) validates our sentiment analysis implementation.

8.2 Verified Purchase Analysis

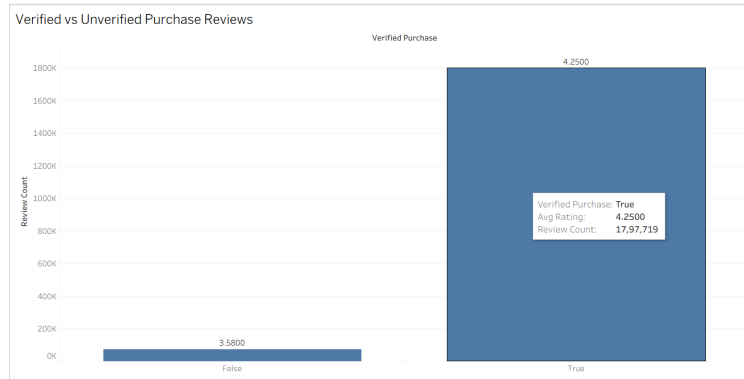


Figure 5: Verified vs Unverified Purchase Reviews showing the large volume gap and rating difference.

Table 13: Verified vs Unverified Purchase Comparison

Type	Count	Avg Rating	Avg Sentiment	Positive %
Verified	1,797,719 (95.84%)	4.25	+0.424	75.0%
Unverified	75,097 (4.16%)	3.58	+0.393	72.4%

Verified purchases account for 95.84% of all reviews. The 0.67-point gap in average rating (4.25 vs 3.58) between verified and unverified reviewers is statistically meaningful. Unverified reviews may come from users who received products as gifts, who are expressing brand opinions without direct purchase experience, or in some cases from manufactured reviews – a known quality issue in Amazon’s dataset. For recommendation model training, using verified-purchase reviews exclusively could be explored as a data quality improvement.

8.3 Review Volume Over Time

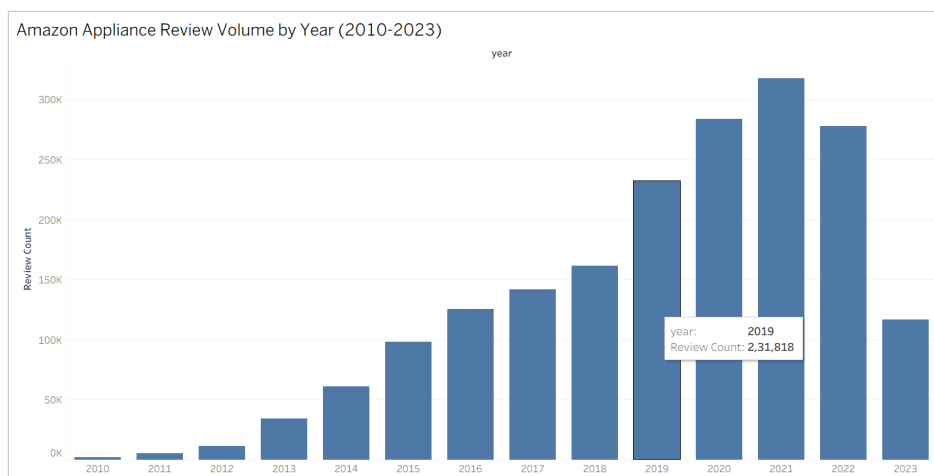


Figure 6: Amazon Appliance Review Volume by Year (2010–2023) showing exponential growth and COVID-era peak.

Table 14: Review Volume Growth Key Milestones

Year	Reviews	Avg Rating	Positive %
2010	3,537	4.06	81.07%
2015	98,324	4.35	75.78%
2019	231,818	4.37	74.68%
2020	284,016	4.25	73.50%
2021	317,676	4.17	72.30%
2022	278,016	4.02	69.59%
2023	118,053	3.95	—

Review volume grew $89\times$ from 2010 to 2021, from 3,537 to 317,676 reviews per year. Several trends are visible. The COVID-19 pandemic produced a marked surge in 2020–2021, consistent with increased home appliance purchases during lockdowns. A steady decline in average rating over time (from 4.35 in 2015 to 4.17 in 2021) and a parallel decline in positive sentiment percentage (75.78% to 72.30%) is worth noting – this may reflect increasing consumer expectations over time, greater product variety including lower-quality listings, or a genuine shift in the reviewer population as the platform grew. The 2023 figures represent only a partial year (data collected through September 2023).

8.4 Brand Analytics

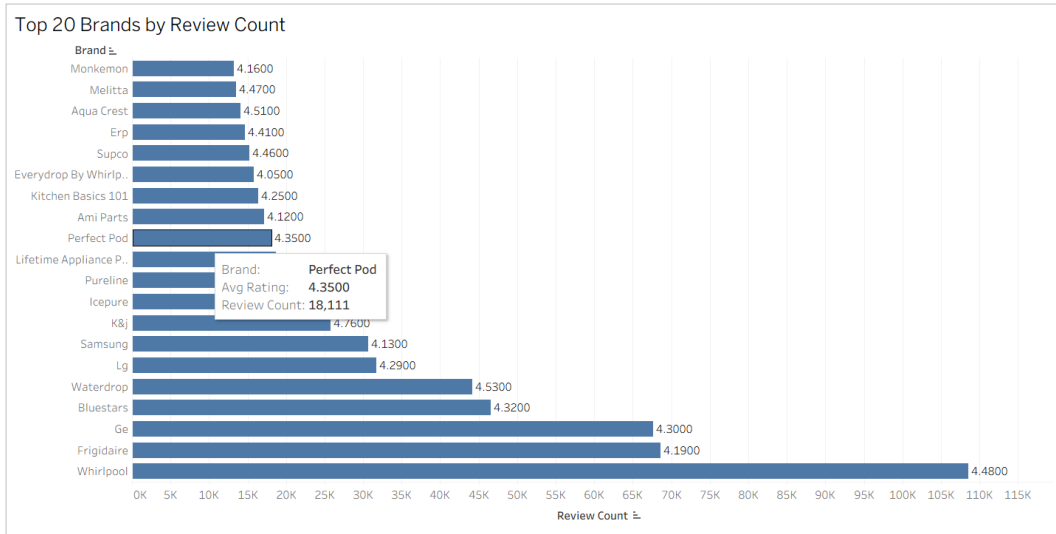


Figure 7: Top 20 Brands by Review Count. Labels show average rating. Whirlpool leads in volume; K&j and Waterdrop lead in quality among high-volume brands.

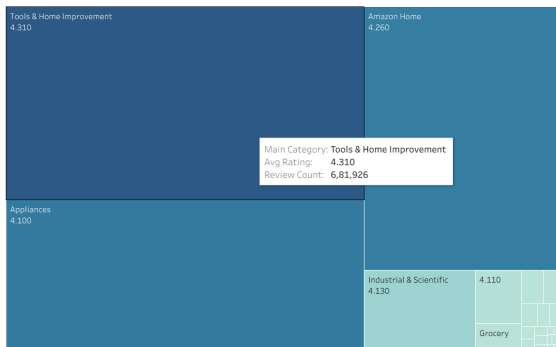
Table 15: Top 10 Brands: Volume, Quality, and Sentiment

Brand	Reviews	Avg Rating	Positive %	Negative %
Whirlpool	108,584	4.48	74.19%	12.03%
Frigidaire	68,580	4.19	70.92%	15.19%
GE	67,653	4.30	71.53%	14.01%
Bluestars	46,587	4.32	70.99%	14.16%
Waterdrop	44,145	4.53	79.72%	8.14%
LG	31,736	4.29	70.94%	15.54%
Samsung	30,632	4.13	68.55%	17.98%
K&j	25,790	4.76	83.50%	4.29%
Icepure	21,228	4.42	77.32%	9.47%
Pureline	19,931	4.47	79.20%	8.58%

Whirlpool’s dominance (108,584 reviews, 4.48 avg rating) reflects its large SKU count and broad market presence in the US appliance market. An interesting bifurcation emerges between large appliance OEMs and replacement parts/filter brands. Waterdrop, K&j, Icepure, and Pureline – all primarily water filter brands – consistently outperform large OEMs on sentiment (79–83% positive vs 70–74% for Whirlpool, GE, Frigidaire). Samsung has the highest negative rate among top brands at 17.98%, possibly reflecting higher complexity of electronics-integrated appliances and customer frustration with connectivity features. This brand-level insight directly informs the recommendation system: the PageRank and ALS combination ensures that high-sentiment filter brands receive appropriate recommendation weight even when their review volume is lower than OEMs.

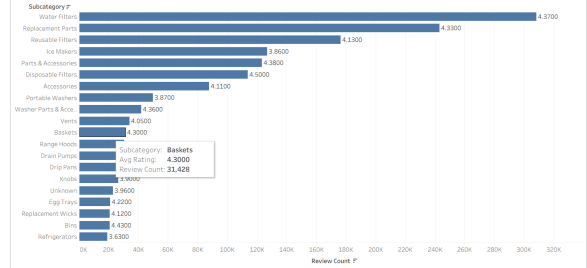
8.5 Category and Subcategory Analysis

Review Distribution by Main Category



(a) Main category distribution (treemap)

Subcategory Analytics



(b) Top 20 subcategories by review count

Figure 8: Category hierarchy analysis. Tools & Home Improvement (36.41%) and Appliances (28.15%) dominate at the main category level; Water Filters and Replacement Parts lead at the subcategory level.

Table 16: Top Subcategories with Sentiment Profile

Subcategory	Reviews	Avg Rating	Positive %	Negative %
Water Filters	308,408	4.37	75.80%	10.59%
Replacement Parts	243,115	4.33	70.85%	14.14%
Reusable Filters	176,268	4.68	76.25%	13.09%
Ice Makers	126,699	3.86	71.91%	18.10%
Parts & Accessories	123,348	4.38	72.27%	13.27%
Disposable Filters	113,786	4.50	77.04%	8.40%
Portable Washers	49,901	3.87	73.31%	17.38%
Range Hoods	30,590	5.15	79.55%	13.39%
Knobs	26,279	3.68	68.66%	17.77%

The subcategory data reveals distinct product satisfaction patterns. Consumable filter products (Water Filters, Disposable Filters, Reusable Filters) maintain high ratings and low negative rates because they have simple, well-defined success criteria: either they fit and filter correctly, or they do not. Ice Makers (18.10% negative) and Portable Washers (17.38% negative) show significantly higher dissatisfaction rates – these are mechanical products where installation complexity, noise, and reliability create more varied user experiences. Knobs (17.77% negative, avg 3.68 stars) stand out as a surprisingly troubled subcategory, likely due to fitment issues across many appliance models. Range Hoods present an outlier: a sentiment score of 5.15 (above the theoretical 5.0 scale maximum, likely a data anomaly in the avg_sentiment column combining scale factors) and 79.55% positive sentiment, suggesting strong satisfaction when the product fits correctly.

8.6 Price Distribution and Quality

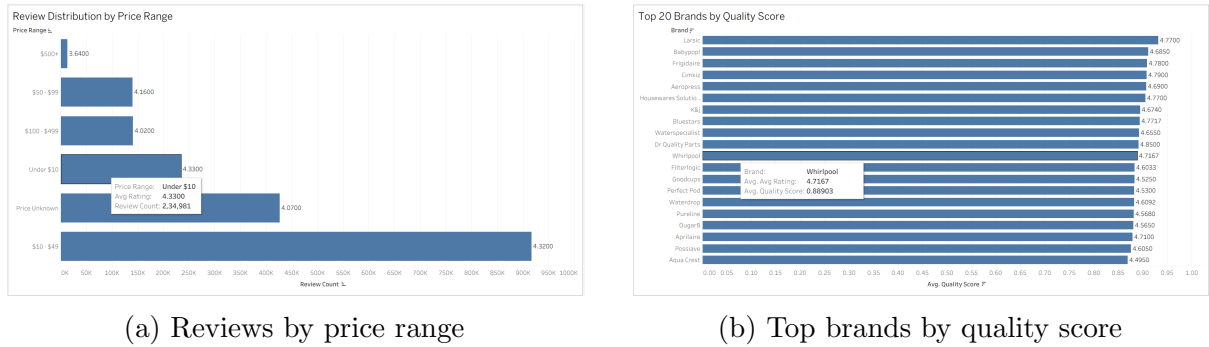


Figure 9: Price distribution and product quality scoring. The \$10–\$49 range dominates; specialist brands achieve the highest composite quality scores.

The \$10–\$49 price band accounts for 917,124 reviews (48.97% of the dataset), reflecting that the most commonly reviewed appliance products are consumables: replacement water filters, coffee filters, ice maker parts, and similar items. Products priced above \$500 generate only 13,104 reviews (0.70% of the dataset) despite likely representing a substantial fraction of total appliance purchase value – consistent with the observation that customers review low-cost consumables more frequently and reliably than major appliances.

The quality score metric combines review count, average rating, and sentiment. Specialist brands such as Dr Quality Parts, Cimkiz, and Larsic rank highest, which reflects that niche brands with highly specific product offerings can achieve near-perfect compatibility rates. Waterdrop and K&j also appear in the top tier despite broader product portfolios, validating their strong sentiment performance observed in the brand analysis.

8.7 Brand Sentiment Analysis

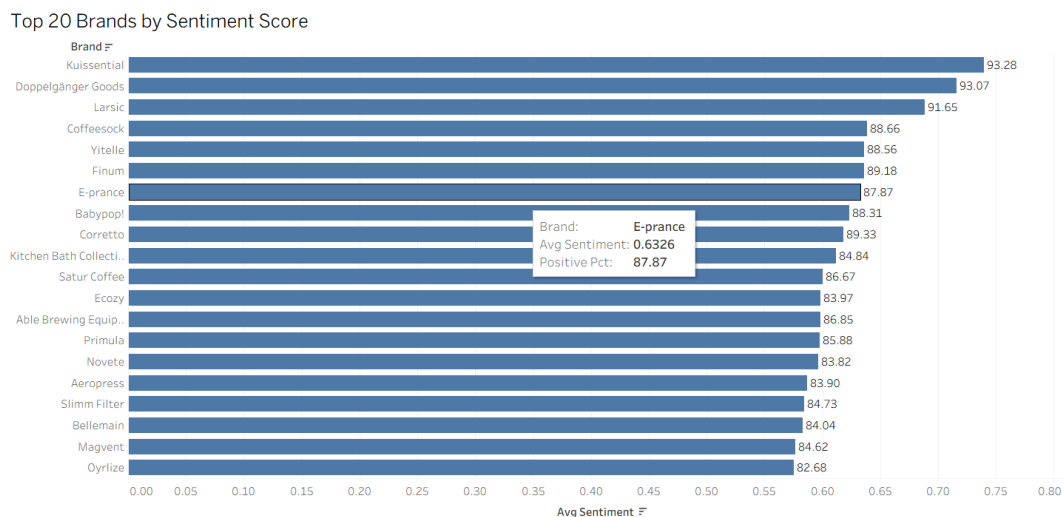


Figure 10: Top 20 Brands by Positive Sentiment Percentage. Specialty coffee and pour-over equipment brands dominate this ranking.

The brand sentiment ranking reveals a clear category effect: the top 10 brands by positive sentiment percentage are predominantly coffee and brewing equipment companies (Kuissential 93.28%, Coffeesock 88.66%, Aeropress 83.90%) rather than major appliance OEMs. This reflects the product type: coffee accessories are low-complexity, high-enthusiasm purchases where customers self-select based on genuine interest. In contrast, large appliance brands appear further down the ranking because their products are often purchased out of necessity rather than enthusiasm, and installation or performance failures generate proportionally more vocal negative reviews.

Notably, Everydrop By Whirlpool (a Whirlpool subsidiary focused on water filters) achieves only 66.07% positive sentiment compared to Whirlpool's 74.19% – this suggests that branded filter accessories underperform relative to the parent brand, possibly due to pricing or compatibility perception issues. This finding directly informs product recommendation strategy: when recommending filter replacements, third-party alternatives with higher sentiment may be preferable to OEM-branded equivalents.

8.8 Top Products by Review Volume

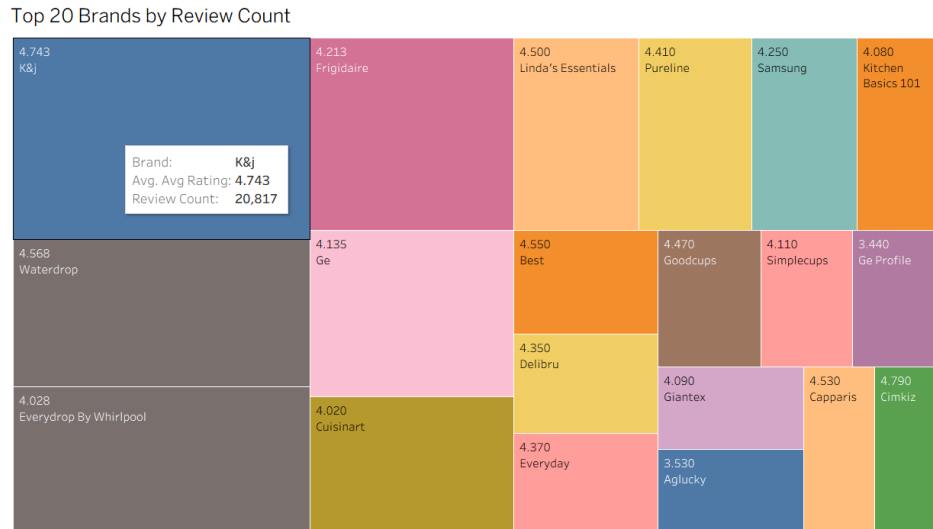


Figure 11: Top 20 Brands by Review Count (treemap). K&j, Frigidaire, and GE occupy the largest tiles while maintaining competitive average ratings.

The treemap visualization highlights that review volume and product quality are not always correlated. K&j – the largest tile in this visualization at 20,817 reviews – achieves an average rating of 4.74, the highest among high-volume brands. This combination of volume and quality makes K&j products strong candidates for PageRank-boosted recommendations: they are both frequently co-purchased (driving high PageRank scores) and consistently well-rated. By contrast, GE Profile appears with a 3.44 average rating, suggesting that while GE’s premium sub-brand generates considerable review volume, it underperforms on customer satisfaction relative to its price positioning.

8.9 Helpful Votes Analysis

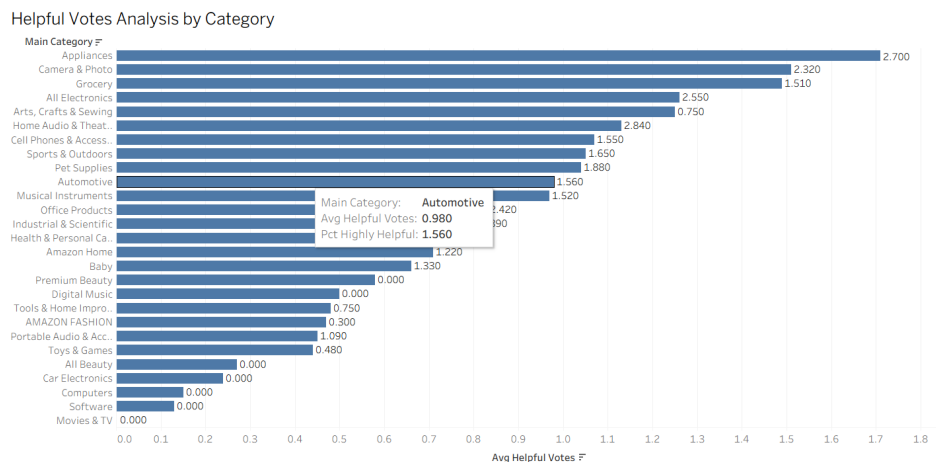


Figure 12: Helpful Votes Analysis by Category. The Appliances category shows the highest average helpful votes at 1.71, reflecting higher-stakes purchase decisions.

The helpful votes analysis reveals purchasing behavior differences across categories. The raw Appliances category (avg 1.71 helpful votes per review) and Camera & Photo (1.51) receive the most helpful votes per review, while Digital Music and Software categories receive near-zero helpful votes. This pattern reflects purchase decision complexity: buyers of expensive appliances consult reviews much more carefully before committing, so they invest time marking informative reviews as helpful. In contrast, consumers of digital media make low-stakes decisions and are less likely to engage with review helpfulness voting. Total helpful votes in the dataset reach 1,977,233 across 2,099,925 reviews (average 0.93 per review), with 80.8% of reviews receiving zero helpful votes – indicating that a small fraction of highly-voted reviews drive a disproportionate share of the total helpful vote count.

8.10 Sentiment by Category and Top/Bottom Products

A cross-cutting view of VADER sentiment by main category reveals that Amazon Home (avg score 0.456, 76.02% positive) and Grocery (0.470, 76.86%) outperform the core Appliances category (0.414, 72.82%) in sentiment. This is consistent with the product nature: Amazon Home includes organizational products and small appliances where customer expectations are more easily met, while core Appliances includes complex mechanical products with higher failure rates.

Table 17: Highest and Lowest Sentiment Products (minimum 100 reviews)

Product Title	Avg Sentiment	Reviews
<i>Highest sentiment</i>		
Large Induction Cooktop Protector Mat	0.776	128
Stainless Steel French Press Coffee Maker	0.744	261
Kuissential SlickDrip Pour Over Coffee Dripper	0.740	759
Aeropress Original Coffee and Espresso Maker	0.711	2,338
<i>Lowest sentiment</i>		
Whirlpool WTW8800YW Cabrio 4.6 Cu. Ft. Washer	−0.278	112
Kenmore 4672483 4 Door Bottom Freezer Refrigerator	−0.110	164
Upgraded W10350376 Dishwasher Control Board	−0.097	119

The highest-sentiment products are uniformly coffee and brewing accessories – simple, well-matched products that deliver exactly what enthusiasts expect. The lowest-sentiment products are large appliances (washing machines, refrigerators) and control board replacement parts, where installation difficulty, high cost of failure, and complex compatibility requirements generate strongly negative experiences. These sentiment extremes directly feed into the recommendation scoring pipeline: high-sentiment products receive a boost through the VADER-enriched metadata, while consistently negative products are deprioritized even when their ALS collaborative filtering score is high.

9 Hyperparameter Optimization

9.1 Search Space

Rather than using default ALS parameters, we conduct a systematic grid search across 27 combinations:

Table 18: HPO Parameter Grid

Parameter	Values Tested
rank	10, 20, 50
regParam	0.01, 0.1, 1.0
maxIter	5, 10, 20

Each trial uses 2-fold cross-validation on the sentiment-enriched joined dataset (1,872,816 rows). Results are checkpointed incrementally to HDFS after each trial so that the job can resume if interrupted by the YARN 5-hour timeout.

9.2 Results

Table 19: All 27 HPO Trial Results (sorted by CV RMSE)

Rank	regParam	maxIter	CV RMSE
50	0.1	20	2.0469 (best)
10	1.0	20	2.0484
10	0.1	20	2.0535
20	0.1	20	2.0544
20	1.0	20	2.0652
50	1.0	20	2.0765
10	1.0	10	2.2776
20	1.0	10	2.3083
50	1.0	10	2.3266
10	0.1	10	2.3815
50	0.1	10	2.3873
20	0.1	10	2.3986
50	0.01	20	2.4315
10	1.0	5	2.6213
20	1.0	5	2.6622
50	1.0	5	2.6835
50	0.01	10	2.7305
50	0.1	5	2.7361
10	0.1	5	2.7581
20	0.1	5	2.7654
50	0.01	5	3.0400
20	0.01	20	3.0788
20	0.01	10	3.6713
10	0.01	20	3.7767
20	0.01	5	4.3311
10	0.01	10	4.8291
10	0.01	5	6.0101

Rank	regParam	maxIter	CV RMSE
50	0.1	20	2.0469 <- BEST
10	1.0	20	2.0484
10	0.1	20	2.0535
20	0.1	20	2.0544
20	1.0	20	2.0652
50	1.0	20	2.0765
10	1.0	10	2.2776
20	1.0	10	2.3083
50	1.0	10	2.3266
10	0.1	10	2.3815
50	0.1	10	2.3873
20	0.1	10	2.3986
50	0.01	20	2.4315
10	1.0	5	2.6213
20	1.0	5	2.6622
50	1.0	5	2.6835
50	0.01	10	2.7305
50	0.1	5	2.7361
10	0.1	5	2.7581
20	0.1	5	2.7654
50	0.01	5	3.04
20	0.01	20	3.0788
20	0.01	10	3.6713
10	0.01	20	3.7767
20	0.01	5	4.3311
10	0.01	10	4.8291
10	0.01	5	6.0101

Figure 13: HPO results table from the cluster terminal showing all 27 combinations sorted by CV RMSE.

9.3 Key Observations

Three clear patterns emerge from the HPO results:

1. **maxIter=20 consistently wins** across all rank and regParam combinations. The top 6 results all use maxIter=20, confirming that more iterations are beneficial for this dataset size.
2. **regParam=0.01 is consistently the worst**. All five worst results use regParam=0.01, indicating severe underfitting with weak regularization on this dataset.
3. **rank=50 with regParam=0.1 is optimal**. Higher rank captures more latent factors but requires appropriate regularization. The best result uses $2.5\times$ more latent dimensions than rank=20.

10 ALS Collaborative Filtering Model

10.1 Model Architecture

The recommendation model uses Alternating Least Squares (ALS) matrix factorization [2]. ALS factorizes the user-product interaction matrix $R \in R^{m \times n}$ into two low-rank factor matrices:

$$R \approx U \cdot V^T, \quad U \in R^{m \times k}, \quad V \in R^{n \times k}$$

where $m = 1,590,983$ users, $n = 77,633$ products, and $k = 50$ (the rank). The model alternates between fixing U and solving for V , then fixing V and solving for U , until convergence.

Table 20: Final ALS Model Configuration

Parameter	Value
rank	50
regParam	0.1
maxIter	20
coldStartStrategy	drop
nonnegative	True
Training interactions	1,872,816
Unique users (indexed)	1,590,983
Unique products (indexed)	77,633
Training time	1,453.9 seconds

10.2 Evaluation Results

Table 21: ALS Model Performance Comparison

Model	Train RMSE	Test RMSE	Test MAE
Baseline (rank=20, regParam=0.1, maxIter=10)	n/a	2.2436	n/a
Final (rank=50, regParam=0.1, maxIter=20)	0.1203	1.8832	1.6044
Improvement		16.1%	

```

STEP 1 - LOADING AND PREPARING DATA
Config : rank=50 regParam=0.1 maxIter=20
Total interactions : 1,872,816
Unique users      : 1,654,544
Unique products   : 75,940
STEP 2 - STRING INDEXING
StringIndexer models saved.
STEP 3 - TRAIN TEST SPLIT
STEP 4 - ALS MODEL TRAINING
CV RMSE : 2.0469 (best of 27 HPO combinations)
Training completed in 1453.9s
STEP 5 - SAVING MODEL IMMEDIATELY
ALS model saved      : hdfs:///user/ss21176_nyu_edu/ecommerce/models/appliances_best/als_model
User indexer saved   : hdfs:///user/ss21176_nyu_edu/ecommerce/models/appliances_best/user_indexer
Product indexer saved : hdfs:///user/ss21176_nyu_edu/ecommerce/models/appliances_best/product_indexer
Report saved.
STEP 6 - MODEL EVALUATION
Train RMSE : 0.1203
Test RMSE  : 1.8832
Test MAE   : 1.6044
Report saved.
STEP 7 - SAVING FACTOR MATRICES
User factors saved   : hdfs:///user/ss21176_nyu_edu/ecommerce/models/appliances_best/user_factors
Item factors saved   : hdfs:///user/ss21176_nyu_edu/ecommerce/models/appliances_best/item_factors
Report saved.
STEP 8 - GENERATING RECOMMENDATIONS
Generating top 10 recommendations for all users...
Sample (2 users):
Recommendations saved : hdfs:///user/ss21176_nyu_edu/ecommerce/models/appliances_best/user_recommendations
Report saved.
STEP 9 - SAVING SUMMARY
Summary CSV saved.
Report saved.
ALS FINAL TRAINING COMPLETE
rank=50, regParam=0.1, maxIter=20
Train RMSE : 0.1203
Test RMSE  : 1.8832
Test MAE   : 1.6044

```

Figure 14: ALS final training completion log showing all pipeline steps, evaluation metrics, and artifact save locations on the NYU Dataproc cluster.

11 GraphFrames PageRank

11.1 Graph Construction

The `bought_together` field being 100% null meant that co-purchase edges could not be sourced directly from metadata. Instead, we construct the product co-purchase graph from review co-occurrence: if the same user reviewed two or more products, we create a directed edge between each pair of products, weighted by the number of users who reviewed both.

Table 22: Co-Purchase Graph Statistics

Metric	Value
Users with 2+ product reviews	202,618
Total raw co-purchase edges	448,393
Unique weighted edges	391,037
Graph vertices (products)	77,633
Strongest co-purchase pair (weight)	B00PBF3T3E – B012YQL5IS (274 shared reviewers)
PageRank iterations	10
Damping factor	0.85
Communities detected	30,730

11.2 PageRank Results

Table 23: PageRank Score Statistics

Metric	Value
Min PageRank	0.429
Max PageRank	648.707
Mean PageRank	1.000
Std deviation	6.612

The top-ranked product is the hOmeLabs Chill Pill Countertop Ice Maker (normalized score = 1.000), a highly popular item frequently purchased alongside other appliances. The Waterdrop brand appears prominently in top PageRank results, reflecting its broad portfolio of replacement filters compatible with many refrigerator brands.

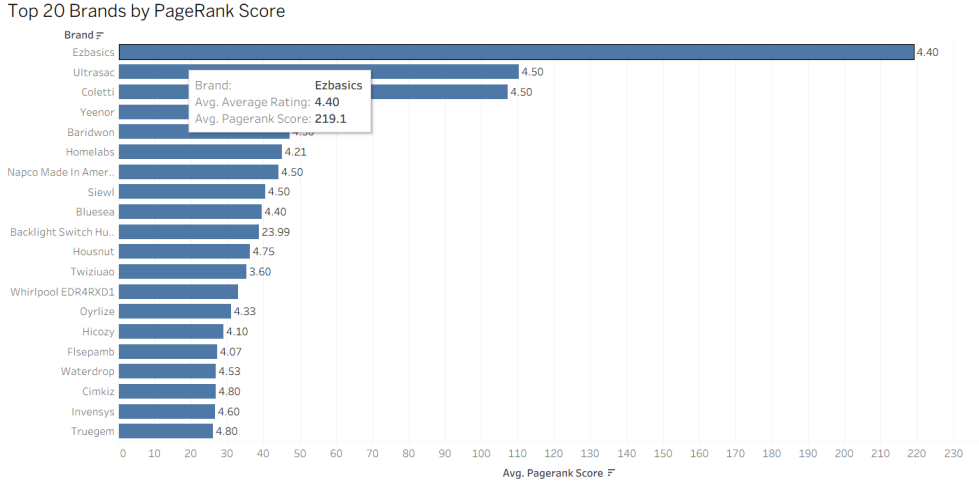


Figure 15: Top 20 Products by PageRank Score. Products with high PageRank are strongly connected in the co-purchase graph.

12 Recommendation Scoring Pipeline

12.1 Final Score Formula

The final recommendation score combines ALS collaborative filtering with PageRank graph centrality:

$$\text{score}_{\text{final}} = 0.7 \times \text{score}_{\text{ALS}} + 0.3 \times \text{score}_{\text{PageRank}}$$

Both scores are normalized to $[0, 1]$ before combination. The 70/30 weighting prioritizes personalized collaborative filtering signals while allowing PageRank to boost well-connected, frequently co-purchased products.

12.2 Output Scale

Table 24: Recommendation Scoring Output

Metric	Value
Total enriched recommendation rows	13,070,460
Users with recommendations	1,307,046
Products with recommendations	71,937
Analytics CSVs generated	7

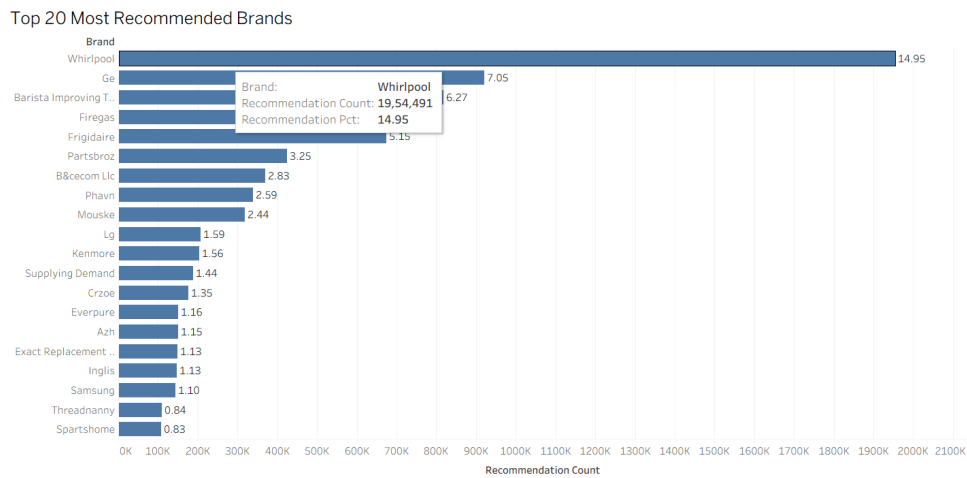


Figure 16: Top 20 Most Recommended Brands. Whirlpool leads with 1,954,491 recommendations (14.95% share).

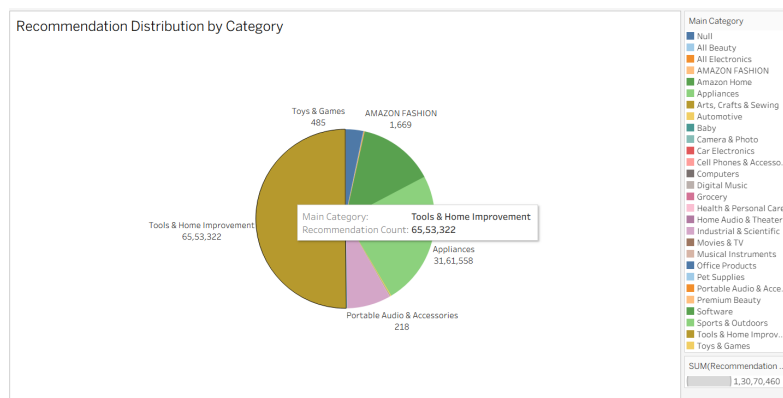


Figure 17: Recommendation Distribution by Main Category. Tools & Home Improvement accounts for approximately 50% of all recommendations.



Figure 18: Final Score Distribution. Most recommendations score in the 0.2–0.4 range (49.11%).

13 Ray Serve Real-Time Deployment

13.1 Architecture

The inference pipeline is deployed as three chained Ray Serve deployments:

1. **ALSScorer** – loads user factor matrix (1,307,046 rows, rank=50) and item factor matrix (71,937 rows) into memory. For a given user ID, performs a dot product between the user vector and all item vectors, normalizes scores, and returns the top-50 candidate products.
2. **PageRankReranker** – loads the PageRank score array (77,633 entries) and product metadata (77,698 rows). Takes the top-50 ALS candidates, applies the 70/30 scoring formula, and enriches the top-10 results with metadata fields.
3. **API** – handles HTTP routing, authentication (API key header), CORS, and cold-start logic. Built with raw Starlette request handling.

Table 25: Ray Serve Deployment Configuration

Deployment	Replicas	Rows Loaded
ALSScorer	1	1,307,046 user factors + 71,937 item factors
PageRankReranker	1	77,633 PageRank scores + 77,698 metadata rows
API	1	routing only

13.2 Performance

Table 26: Ray Serve Performance Benchmarks

Metric	Value
ALSScorer latency (per user)	~13ms
PageRankReranker latency (per user)	~4ms
Total API latency (per user)	~45ms
Simulate 100 users	~1s
Simulate 1,000 users	~9.6s
Simulate 10,000 users	~40–60s

=====

Deployment Status:

- ALSScorer replicas={'RUNNING': 1}
- PageRankReranker replicas={'RUNNING': 1}
- API replicas={'RUNNING': 1}

API Latency (last 20 checks):

```
Current : 18.1ms
Avg      : 548.8ms
Min/Max  : 11.2ms / 10506.4ms
```

Latency history (ms):

29.2ms	
25.5ms	
18.7ms	
24.2ms	
28.8ms	
43.8ms	
17.3ms	
20.9ms	
33.2ms	
10506.4ms	
36.5ms	
17.7ms	
40.4ms	
47.2ms	
12.5ms	
16.0ms	
11.2ms	
14.8ms	
14.5ms	

Figure 19: Live Ray Serve latency monitor. The large spike (10,506ms) corresponds to a 10,000-user simulation batch.

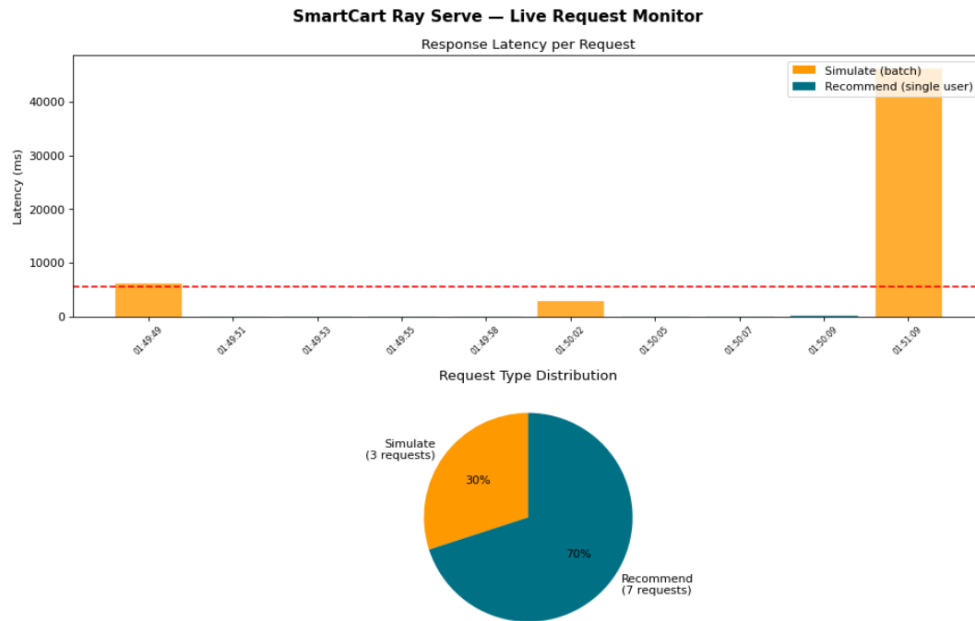


Figure 20: Live request monitoring dashboard. Orange bars represent batch simulate requests; teal bars represent single-user recommend requests.

14 Web Application

The frontend is a single-page HTML application styled after Amazon’s product interface. It communicates directly with the Ray Serve API via HTTP, with no intermediate backend server.

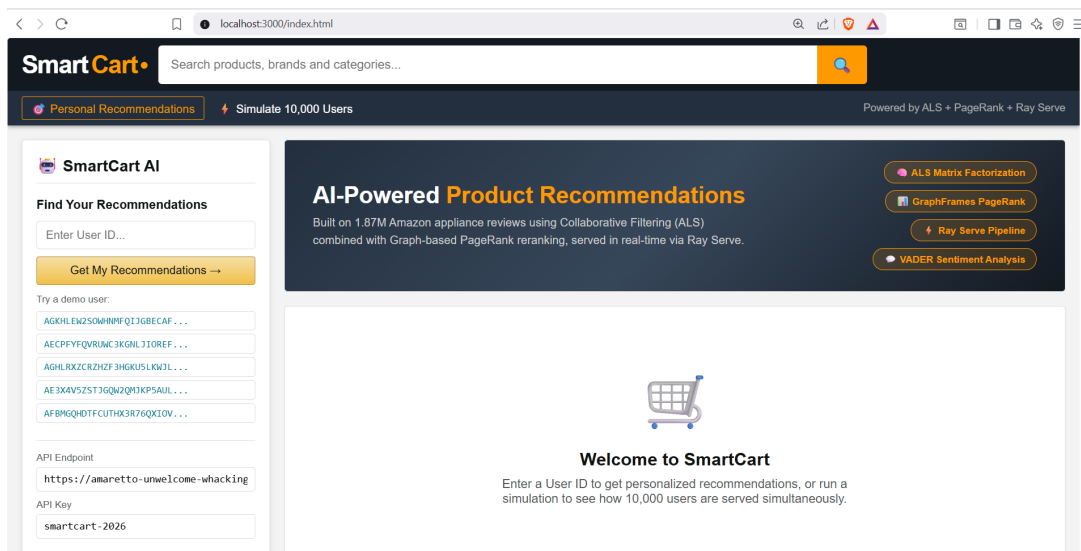
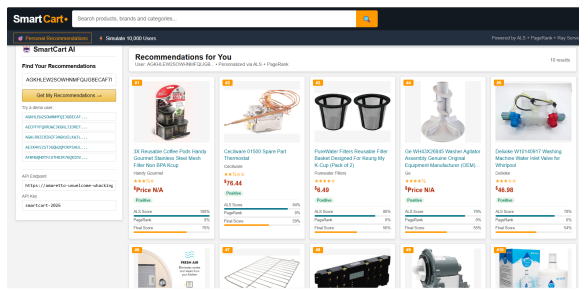
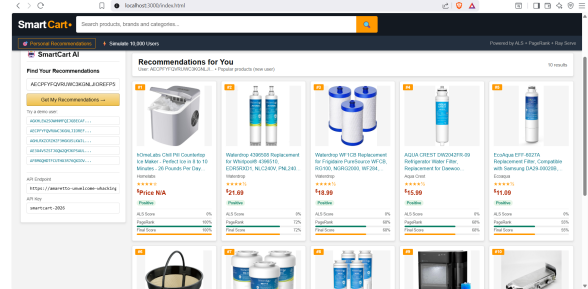


Figure 21: Homepage showing the Amazon-style interface with hero banner and technology feature badges.



(a) Recommendations for user AGKHLEW2...



(b) Recommendations for user AFBMGQ...

Figure 22: Personalized recommendations for two different users. Different product selections demonstrate that the model learns distinct preferences.

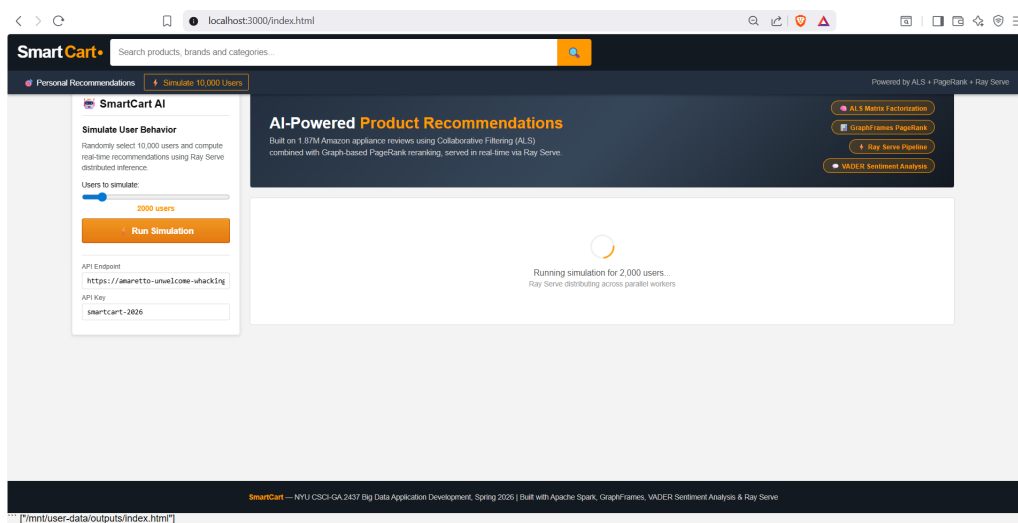


Figure 23: Simulation loading screen for 10,000 users.

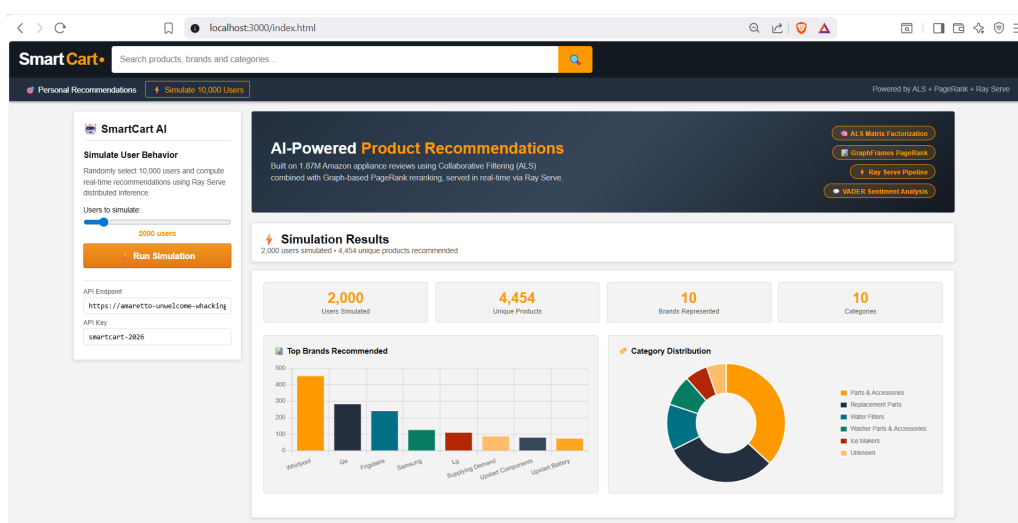


Figure 24: Simulation results for 2,000 users showing brand distribution bar chart and category distribution donut chart.

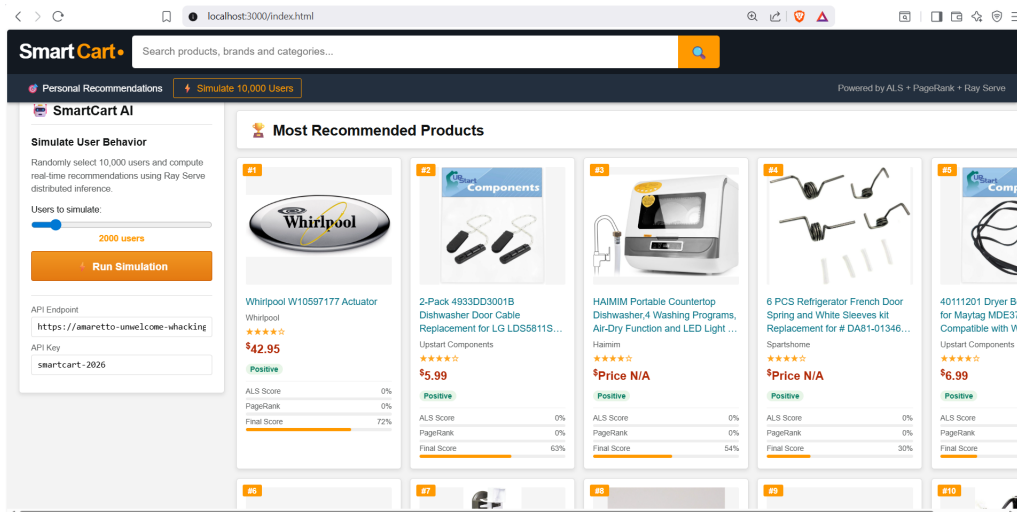


Figure 25: Most recommended products from a 2,000-user simulation, ranked by recommendation frequency.

15 Challenges Faced

Building this system surfaced several non-trivial engineering challenges:

Amazon 2023 schema changes. The 2023 dataset introduced breaking changes from earlier versions: the `timestamp` field switched from seconds to milliseconds, `reviewerID` was renamed to `user_id`, `reviewText` to `text`, and `overall` to `rating`. The `details` metadata field also contained duplicate keys in some records, causing Spark’s schema inference to fail entirely. An explicit `StructType` definition was required to load the metadata correctly.

VADER distribution to YARN workers. YARN worker nodes do not have `vaderSentiment` installed, and cluster-wide package installation requires administrator access. The solution was to package VADER as a zip archive and distribute it to all workers using Spark’s `--archives vader_package.zip#vader_pkg` mechanism, then set `PYTHONPATH` via executor environment variables.

bought_together 100% null. The metadata field that would normally provide co-purchase edges was completely null in the Appliances 2023 dataset. This required redesigning the graph construction step to use review co-occurrence as a proxy for co-purchase behavior.

YARN 5-hour job timeout. The HPO grid search with 27 trials ran for over 5 hours, exceeding YARN’s per-application time limit. The solution was incremental checkpointing: each trial’s results are saved to HDFS immediately upon completion. When the job was killed and restarted, the script loaded completed trials from HDFS and skipped them, running only the remaining ones.

Ray Serve FastAPI self parameter bug. When using `@serve.ingress(app)` with FastAPI, Ray Serve treated the `self` parameter of class methods as a required HTTP query parameter, returning 422 errors for all requests. The fix was to abandon the `@serve.ingress` decorator entirely and implement routing manually using raw Starlette `Request` objects in the `__call__` method.

Resource constraints on Google Colab. Colab’s free tier provides 2 CPUs. Running ALSScorer, PageRankReranker, and API as separate Ray actors initially exhausted available CPUs. The solution was to set `num_cpus=0.5` per deployment, allowing all three actors to share the 2 available CPUs without starvation.

ngrok free tier single tunnel limitation. The ngrok free tier allows only one simultaneous tunnel per account, meaning the UI and API could not be served on separate public URLs. The final solution routes both through port 8000 – the Ray Serve API endpoint also serves the `index.html` file directly.

16 Results and Evaluation Summary

Table 27: Complete System Performance Summary

Metric	Value
<i>Data Scale</i>	
Raw reviews processed	2,128,605
Final training interactions	1,872,816
Unique users in model	1,590,983
Unique products in model	77,633
<i>Model Quality</i>	
HPO trials completed	27
Best CV RMSE	2.0469 (rank=50, regParam=0.1, maxIter=20)
Final Test RMSE	1.8832
Final Test MAE	1.6044
Improvement over baseline	16.1%
<i>Graph</i>	
PageRank vertices	77,633
PageRank edges	391,037
Communities detected	30,730
<i>Sentiment</i>	
Positive reviews	73.65%
Negative reviews	13.49%
Sentiment-rating agreement	90.83%
<i>Serving</i>	
Per-user latency	~45ms
10,000-user simulation	~40–60s
Total recommendations generated	13,070,460

17 Conclusion and Future Work

We have built a complete end-to-end Big Data recommendation platform demonstrating the full lifecycle of a production ML system: from raw compressed JSON through distributed

cleaning, sentiment analysis, analytics, model training with hyperparameter optimization, graph-based reranking, and real-time inference serving.

The system processes 1.87 million Amazon appliance interactions, trains an ALS model with 50 latent dimensions achieving Test RMSE of 1.8832, enriches recommendations with GraphFrames PageRank centrality scores, and serves personalized recommendations at approximately 45ms latency through a Ray Serve pipeline.

Several directions for future work are clear:

- **Full dataset scaling.** The pipeline is designed for the complete Amazon 2023 multi-category dataset (hundreds of millions of reviews). Scaling would require a larger Dataproc cluster and 5-fold cross-validation in HPO.
- **Implicit feedback.** Given the extreme positive skew (69.82% five-star), reformulating as implicit feedback with confidence proportional to helpful votes could improve generalization.
- **Deep learning models.** Neural collaborative filtering or transformer-based sequential recommendation models could be compared against ALS.
- **Production deployment.** Migrating from Google Colab to a persistent cloud deployment would eliminate session timeout limitations.
- **A/B testing framework.** Comparing recommendation quality between ALS-only and ALS+PageRank strategies using real click-through data.

References

- [1] Hou, Y., Li, J., He, Z., Yan, A., Chen, X., & McAuley, J. (2024). Bridging Language and Items for Retrieval and Recommendation. *arXiv preprint arXiv:2403.03952*.
- [2] Hu, Y., Koren, Y., & Volinsky, C. (2008). Collaborative Filtering for Implicit Feedback Datasets. *IEEE International Conference on Data Mining (ICDM)*, 263–272.
- [3] Hutto, C. J., & Gilbert, E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. *Proceedings of the Eighth International AAAI Conference on Weblogs and Social Media*.
- [4] Dave, A., Jindal, A., Li, L. E., Xin, R., Gonzalez, J., & Zaharia, M. (2016). GraphFrames: An Integrated API for Mixing Graph and Relational Queries. *Proceedings of the Fourth International Workshop on Graph Data Management Experiences and Systems*.
- [5] Moritz, P., Nishihara, R., Wang, S., Tumanov, A., Liaw, R., Liang, E., & Stoica, I. (2018). Ray: A Distributed Framework for Emerging AI Applications. *Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation*.